8. Energies, capacities and subsets of finite measure

Energies

One of the basic themes of this book is the study of geometric properties of general Radon measures μ on $\mathbf{R}^n$. Conditions we often impose on them guarantee that not too much measure is concentrated on small regions. This can be expressed for example by the growth condition with some positive numbers s and c,

$$\mu(B(x,r)) \leq cr^s \quad \text{for } x \in \mathbf{R}^n,\ 0 < r < \infty, \tag{8.1}$$

or by the finiteness of *t-energy* $I_t(\mu)$,

$$I_t(\mu) = \iint |x-y|^{-t}\, d\mu x\, d\mu y < \infty. \tag{8.2}$$

We shall see that the conditions (8.1) and (8.2) are very closely related to each other and also to the Hausdorff measures and dimension.

To get some feeling what (8.1) and (8.2) mean, consider $\mu = \mathcal{L}^1 \llcorner [0,1]$. Then (8.1) holds if and only if $s \leq 1$, and (8.2) holds if and only if $t < 1$. This is of course very easy to see. It takes a little more work to show that for any non-zero Radon measure μ on $[0,1]$, (8.1) can hold only if $s \leq 1$, and (8.2) can hold only if $t < 1$. Thus in both cases the range of the possible parameters s and t is bounded from above by 1, which is also the dimension of $[0,1]$. This is no coincidence, and we come to that in greater generality soon.

Let us compare first the conditions (8.1) and (8.2). Using Theorem 1.15,

$$\begin{aligned}\int |x-y|^{-t}\, d\mu y &= \int_0^\infty \mu\big(\{y : |x-y|^{-t} \geq u\}\big)\, du \\ &= \int_0^\infty \mu(B(x, u^{-1/t}))\, du = t\int_0^\infty r^{-t-1}\mu(B(x,r))\, dr\end{aligned}$$

by a change of variable. If $\mu(\mathbf{R}^n) < \infty$ and if for some $s > t$, $\mu(B(x,r)) \leq cr^s$ for $x \in \mathbf{R}^n$, $r > 0$, then we immediately see that $I_t(\mu) < \infty$. On the other hand if $I_s(\mu) < \infty$, (8.1) need not quite hold, but it holds for a suitable restriction of μ. Namely, assuming $0 < \mu(\mathbf{R}^n) < \infty$, the Borel set

$$A = \Big\{x : \int |x-y|^{-s}\, d\mu y \leq M\Big\}$$

has positive μ measure for some M. If $\nu = \mu \llcorner A$, then

$$r^{-s}\nu(B(x,r)) \le \int_{B(x,r)} |x-y|^{-s}\, d\nu y \le M \quad \text{for } x \in A,\ r > 0.$$

To see that ν really satisfies (8.1), let $x \in \mathbf{R}^n$ and $r > 0$. If $B(x,r) \cap A = \emptyset$, $\nu(B(x,r)) = 0$. If there is $z \in B(x,r) \cap A$, we have by the above

$$r^{-s}\nu(B(x,r)) \le 2^s(2r)^{-s}\nu(B(z,2r)) \le 2^s M.$$

This discussion shows that the two least upper bounds in the next definition agree. For $A \subset \mathbf{R}^n$, let

$$\begin{aligned}\mathcal{M}(A) = \{\mu : \mu \text{ is a Radon measure with compact support,}\\ \operatorname{spt}\mu \subset A \text{ and } 0 < \mu(\mathbf{R}^n) < \infty\}.\end{aligned}$$

8.3. Definition. *The capacitary dimension of a set $A \subset \mathbf{R}^n$ is*

$$\begin{aligned}\dim_c A &= \sup\left\{s : \exists\mu \in \mathcal{M}(A) \text{ with } \mu(B(x,r)) \le r^s \text{ for } x \in \mathbf{R}^n,\, r > 0\right\}\\ &= \sup\left\{t : \exists\mu \in \mathcal{M}(A) \text{ with } I_t(\mu) < \infty\right\}.\end{aligned}$$

Here the supremum is interpreted as 0 if there are no such parameters s or t. For the first this occurs only if $A = \emptyset$. For the second there is no non-zero Radon measure μ on A with $I_t(\mu) < \infty$ for some $t > 0$ if A is finite or countable. There are also uncountable compact sets with this property; in fact, we shall soon see that they are exactly those having Hausdorff dimension zero.

Capacities and Hausdorff measures

We can also arrive at the capacitary dimension through set functions called Riesz capacities.

8.4. Definition. *Let $s > 0$. The (Riesz) s-capacity of a set $A \subset \mathbf{R}^n$ is defined by*

$$C_s(A) = \sup\left\{I_s(\mu)^{-1} : \mu \in \mathcal{M}(A) \text{ with } \mu(\mathbf{R}^n) = 1\right\}$$

with the interpretation $C_s(\emptyset) = 0$.

The following result is merely a restatement of the definitions.

8.5. Theorem. *For $s > 0$ and $A \subset \mathbf{R}^n$,*

$$\dim_c A = \sup\{s : C_s(A) > 0\} = \inf\{s : C_s(A) = 0\}.$$

8.6. Remarks. By a trivial approximation we could drop the requirement that the measures μ have compact support in the definitions of $\dim_c A$ and $C_s(A)$. More generally we could use Borel measures instead of Radon measures. Alternative definitions and relations to potential theory and complex analysis can be found e.g. in Carleson [1], Hayman and Kennedy [1] and Landkof [1].

The capacity C_s is a measure on $\mathbf{R}^n$ although highly non-additive. For example if $0 < s \leq n-2$, then $0 < C_s(B(x,r)) = C_s(U(x,r)) = C_s(S(x,r)) < \infty$ for $x \in \mathbf{R}^n$, $0 < r < \infty$, see Landkof [1, pp. 141 and 163].

Note that $C_s(A) > 0$ if and only if there is $\mu \in \mathcal{M}(A)$ with $I_s(\mu) < \infty$. It is clear from the earlier discussion on $A = [0,1]$ that $C_s([0,1]) > 0$ if and only if $0 < s < 1$, whence $\dim_c[0,1] = 1$. This holds more generally. Suppose $s > 0$, E is $\mathcal{H}^s$ measurable and $0 < \mathcal{H}^s(E) < \infty$. Then by Theorem 6.2, $\Theta^{*s}(E,x) \leq 1$ for $\mathcal{H}^s$ almost all $x \in \mathbf{R}^n$. Using this one finds a restriction μ of $\mathcal{H}^s \llcorner A$ to a suitable subset of E satisfying (8.1). Thus $\dim_c E \geq s$. On the other hand, it follows from the next, rather simple, theorem that also $\dim_c E \leq s$ and so $\dim_c E = s$. Of course, this extends immediately if E has only σ-finite $\mathcal{H}^s$ measure. In particular, $\dim_c \mathbf{R}^n = n$ and $C_s(\mathbf{R}^n) = 0$ for $s > n$.

In the above discussion we saw that $\dim_c E = \dim E$ provided E is $\mathcal{H}^s$ measurable and has positive and σ-finite $\mathcal{H}^s$ measure. One of the main results of this chapter is Theorem 8.9, which says that this holds, at least for Borel sets, without the positivity and finiteness assumptions. The main tool for proving this for closed sets will be Frostman's lemma, Theorem 8.8. For general Borel sets the theory of Suslin sets is required, and we omit that part, see Carleson [1], Hayman and Kennedy [1] or Landkof [1].

Before going on let us make one more trivial observation. In contrast to Hausdorff measures the finiteness of capacities says very little: it is easy to see that any bounded set in $\mathbf{R}^n$ has finite s-capacity for all $s > 0$.

8.7. Theorem. *Let $A \subset \mathbf{R}^n$.*

(1) *If $s > 0$ and $\mathcal{H}^s(A) < \infty$, then $C_s(A) = 0$.*
(2) $\dim_c A \leq \dim A$.

Proof. (1) Suppose $C_s(A) > 0$. Then there is $\mu \in \mathcal{M}(A)$ with $\mu(A) = 1$ and $I_s(\mu) < \infty$. Thus $\int |x-y|^{-s} d\mu y < \infty$ for μ almost all $x \in \mathbf{R}^n$, whence for such x,

$$\lim_{r \downarrow 0} \int_{B(x,r)} |x-y|^{-s}\, d\mu y = 0.$$

Consequently, given $\varepsilon > 0$ there are $B \subset A$ and $\delta > 0$ such that $\mu(B) > 1/2$ and

$$\mu(B(x,r)) \le r^s \int_{B(x,r)} |x-y|^{-s}\, d\mu y \le \varepsilon r^s \quad \text{for } x \in B \text{ and } 0 < r \le \delta.$$

Choose sets $E_1, E_2, \dots$ such that

$$B \subset \bigcup_i E_i, \quad B \cap E_i \neq \emptyset, \quad d(E_i) \le \delta \quad \text{and}$$
$$\sum_i d(E_i)^s \le \mathcal{H}^s(A) + 1.$$

Picking $x_i \in B \cap E_i$ and setting $r_i = d(E_i)$, we have

$$1/2 < \mu(B) \le \sum_i \mu(B(x_i, r_i)) \le \varepsilon \sum_i r_i^s \le \varepsilon(\mathcal{H}^s(A) + 1).$$

Letting $\varepsilon \downarrow 0$ we conclude $\mathcal{H}^s(A) = \infty$, which proves (1).

(2) follows immediately from (1). □

Frostman's lemma in $\mathbf{R}^n$

We shall now prove Frostman's lemma for $\mathcal{F}_\sigma$-sets, that is, countable unions of closed sets. As remarked before, the case of Borel, and more generally, Suslin sets is more difficult and treated in Carleson [1]. At the end of this chapter we shall give a different proof due to Howroyd. Both proofs apply also to generalized Hausdorff measures Λ_h.

8.8. Theorem. *Let B be a Borel set in $\mathbf{R}^n$. Then $\mathcal{H}^s(B) > 0$ if and only if there exists $\mu \in \mathcal{M}(B)$ such that $\mu(B(x,r)) \le r^s$ for $x \in \mathbf{R}^n$ and $r > 0$. Moreover, we can find μ so that $\mu(B) \ge c\mathcal{H}^s_\infty(B)$ where $c > 0$ depends only on n.*

Proof. If such a μ exists, a simple argument as in the proof of Theorem 8.7 gives, even for arbitrary sets $B \subset \mathbf{R}^n$, that $\mathcal{H}^s(B) > 0$.

To prove the converse part for $\mathcal{F}_\sigma$-sets, we may obviously assume that B is compact. Translating B we may also assume that B is contained in some dyadic cube. Since $\mathcal{H}^s(B) > 0$, also $\mathcal{H}^s_\infty(B) > 0$, and so with $b = \mathcal{H}^s_\infty(B)$,

$$\sum_i d(Q_i)^s \geq b,$$

whenever the cubes $Q_1, Q_2, \dots$ cover B.

For $m = 1, 2, \dots$, denote by $\mathcal{D}_m$ the family of dyadic cubes of $\mathbf{R}^n$ with side-length 2^{-m}, recall 5.2. Define a measure μ^m_m on $\mathbf{R}^n$ by requiring that for all $Q \in \mathcal{D}_m$,

$$\begin{aligned} &\mu^m_m \llcorner Q = 2^{-ms}\mathcal{L}^n(Q)^{-1}(\mathcal{L}^n \llcorner Q), && \text{if } B \cap Q \neq \emptyset,\\ &\mu^m_m \llcorner Q = 0, && \text{if } B \cap Q = \emptyset. \end{aligned}$$

Next we modify μ^m_m, defining a measure μ^m_{m-1} by requiring for all $Q \in \mathcal{D}_{m-1}$ that

$$\begin{aligned} &\mu^m_{m-1} \llcorner Q = \mu^m_m \llcorner Q, && \text{if } \mu^m_m(Q) \leq 2^{-(m-1)s},\\ &\mu^m_{m-1} \llcorner Q = 2^{-(m-1)s}\mu^m_m(Q)^{-1}(\mu^m_m \llcorner Q), && \text{if } \mu^m_m(Q) > 2^{-(m-1)s}. \end{aligned}$$

We continue this; μ^m_{m-k-1} is obtained from μ^m_{m-k} in such a way that for $Q \in \mathcal{D}_{m-k-1}$,

$$\begin{aligned} &\mu^m_{m-k-1} \llcorner Q = \lambda(Q)(\mu^m_{m-k} \llcorner Q) \quad \text{where}\\ &\lambda(Q) = \min\left\{1, 2^{-(m-k-1)s}\mu^m_{m-k}(Q)^{-1}\right\}. \end{aligned}$$

We stop as soon as $B \subset Q$ for some $Q \in \mathcal{D}_{m-k_0}$ and then define $\mu^m = \mu^m_{m-k_0}$. Since at no stage did the measure of any dyadic cube increase, μ^m satisfies

$$\mu^m(Q) \leq 2^{-(m-k)s} \quad \text{for } Q \in \mathcal{D}_{m-k},\ k = 0, 1, 2, \dots.$$

Further, it follows from the construction that for every $x \in B$ there are k and $Q \in \mathcal{D}_{m-k}$ such that $x \in Q$ and

$$\mu^m(Q) = 2^{-(m-k)s} = n^{-s/2}d(Q)^s.$$

Picking for each x the largest such Q we obtain disjoint cubes $Q_1, \ldots, Q_k$ such that $B \subset \bigcup_{i=1}^k Q_i$ and

$$\mu^m(\mathbf{R}^n) = \sum_{i=1}^k \mu^m(Q_i) = n^{-s/2} \sum_{i=1}^k d(Q_i)^s \geq n^{-s/2} b.$$

Let $\nu^m = \mu^m(\mathbf{R}^n)^{-1} \mu^m$. Then $\nu^m(\mathbf{R}^n) = 1$ and

$$\nu^m(Q) \leq b^{-1} n^{s/2} 2^{-(m-k)s} \quad \text{for } Q \in \mathcal{D}_{m-k}, \ k = 0, 1, 2, \ldots.$$

By Theorem 1.23 the sequence (ν^m) has a weakly convergent subsequence $\nu^{m_i} \xrightarrow{\mathrm{w}} \nu$. Clearly $\nu \in \mathcal{M}(B)$ (as B is compact) with $\nu(B) = 1$. For any $x \in \mathbf{R}^n$ and $0 < r < \infty$, $B(x, r)$ is contained in the interior U of a union $\bigcup_{i=1}^{2^n} Q_i$ of 2^n cubes $Q \in \mathcal{D}_p$ with $d(Q) = n^{1/2} 2^{-p} \leq 4n^{1/2} r$. Hence for $m \geq p$,

$$\nu^m(U) \leq 2^n b^{-1} n^{s/2} 2^{-ps} \leq 2^{n+2s} b^{-1} n^{s/2} r^s,$$

and so by Theorem 1.24 (2)

$$\nu(B(x, r)) \leq \nu(U) \leq \liminf_{i \to \infty} \nu^{m_i}(U) \leq 2^{n+2s} b^{-1} n^{s/2} r^s.$$

This completes the proof. □

Using Frostman's lemma we can now give more complete information about the relations of Hausdorff measures and capacities of Borel sets. The following theorem is often very useful for the estimation of Hausdorff dimension from below.

8.9. Theorem. *Let A be a Borel set in $\mathbf{R}^n$.*

(1) *If $s > 0$ and $\mathcal{H}^s(A) < \infty$, then $C_s(A) = 0$.*
(2) *If $s > 0$ and $C_s(A) = 0$, then $\mathcal{H}^t(A) = 0$ for $t > s$.*
(3) *$\dim_c A = \dim A$.*

Proof. (1) was already stated in Theorem 8.7 (1). If $\mathcal{H}^t(A) > 0$, Frostman's lemma 8.8 gives $\mu \in \mathcal{M}(A)$ for which $\mu(B(x, r)) \leq r^t$. Then for $0 < s < t$, $I_s(\mu) < \infty$ by the discussion at the beginning of this chapter. Hence $C_s(A) > 0$ and (2) follows. (3) is an immediate consequence of (1) and (2). □

Dimensions of product sets

Frostman's lemma easily gives information about the Hausdorff dimension of cartesian products. We shall also study packing dimension in this connection.

8.10. Theorem. *Let $A \subset \mathbf{R}^m$ and $B \subset \mathbf{R}^n$ be non-empty Borel sets. Then*

(1) $\quad \mathcal{H}^{s+t}(A \times B) > 0$ *provided* $\mathcal{H}^s(A) > 0$ *and* $\mathcal{H}^t(B) > 0$,

(2) $$\dim A + \dim B \leq \dim(A \times B) \leq \dim A + \overline{\dim}_p B$$

and

(3) $$\overline{\dim}_p(A \times B) \leq \overline{\dim}_p A + \overline{\dim}_p B.$$

Proof. To prove the first statement, let $\mu \in \mathcal{M}(A)$ and $\nu \in \mathcal{M}(B)$ be Radon measures given by Frostman's lemma 8.8 such that $\mu(B(x,r)) \leq r^s$ and $\nu(B(y,r)) \leq r^t$ for $x \in \mathbf{R}^m$, $y \in \mathbf{R}^n$ and $r > 0$. Then $\mu \times \nu \in \mathcal{M}(A \times B)$ and

$$\begin{aligned}(\mu \times \nu)\big(B((x,y),r)\big) &\leq (\mu \times \nu)\big(B(x,r) \times B(y,r)\big) \\ &= \mu(B(x,r)) \cdot \nu(B(y,r)) \leq r^{s+t}.\end{aligned}$$

Hence by Frostman's lemma $\mathcal{H}^{s+t}(A \times B) > 0$.

The first inequality in (2) follows easily from (1). Recalling 5.9 one sees that the second inequality is reduced to

(4) $$\dim(A \times B) \leq \dim A + \overline{\dim}_M B$$

for all bounded Borel sets B. To prove this let $\dim A < s$ and $\overline{\dim}_M B < t$ so that $\mathcal{H}^s(A) = 0$ and $\lim_{\varepsilon \downarrow 0} N(B,\varepsilon)\varepsilon^t = 0$. Let $\delta > 0$ be such that $N(B,\varepsilon)(2\varepsilon)^t < 1$ for $0 < \varepsilon < \delta$. We can cover A with sets $E_1, E_2, \dots$ such that $0 < d(E_i) \leq \delta/2$ and $\sum_i d(E_i)^s < 1$. For each i we can then cover B with N_i balls $B_{i,j}$, $j = 1, \dots, N_i$, of diameter $d(E_i)$ such that $N_i d(E_i)^t < 1$. Then the sets $E_i \times B_{i,j}$ together cover $A \times B$. Since $d(E_i \times B_{i,j}) \leq 2d(E_i) \leq \delta$, we have

$$\begin{aligned}\mathcal{H}^{s+t}_\delta(A \times B) &\leq \sum_{i,j} d(E_i \times B_{i,j})^{s+t} \\ &\leq 2^{s+t} \sum_i d(E_i)^s N_i d(E_i)^t \leq 2^{s+t} \sum_i d(E_i)^s < 2^{s+t}.\end{aligned}$$

Letting $\delta \downarrow 0$ we get $\mathcal{H}^{s+t}(A \times B) < \infty$, which yields (4).

We leave the proof of the last inequality as an exercise. □

8.11. Corollary. *If A and B are Borel sets in $\mathbf{R}^n$ and $\dim B = \overline{\dim}_p B$, then*

$$\dim(A \times B) = \dim A + \dim B.$$

8.12. Remarks. (1) Remember that the assumption $\dim B = \overline{\dim}_p B$ holds for example if B is self-similar, recall Corollary 5.8 and the remark following it, or if $0 < \mathcal{H}^t(B) < \infty$ and $\Theta^t_*(B, y) > 0$ for $y \in B$ by Theorem 6.13. In this case the dimension formula of 8.11 was proved by Besicovitch and Moran [1]. The packing dimension part of Theorem 8.10 and its corollary as well as some further inequalities are due to Tricot [2]. A formulation for measures is given in Haase [3], see also Hu and Taylor [1]. Without some extra assumption the formula of 8.11 does not hold even for compact sets A and B; an example has been constructed by Marstrand [2], see also Federer [3, 2.10.29]. In the same paper Marstrand showed that the inequality $\dim A + \dim B \leq \dim(A \times B)$ holds for all sets $A \subset \mathbf{R}^m$ and $B \subset \mathbf{R}^n$, see also Falconer [4, § 5.3]. This is considerably harder since Frostman's lemma is not available. With methods which we shall soon explain, Howroyd [1] has extended the inequality $\dim A + \dim B \leq \dim(A \times B)$ to subsets of arbitrary metric spaces. For a sharp inequality for Hausdorff measures of product sets, see Ernst and Freilich [1]. Buczolich [2] constructed some examples related to product and sum sets.

(2) The above relations between Hausdorff measures and capacities are essentially due to Frostman [1]. They are an indication of the significance of Hausdorff measures and dimension in several areas of analysis. For $n \geq 3$, C_{n-2} is the classical capacity related to potential theory and harmonic functions; in $\mathbf{R}^2$ one has to use a logarithmic kernel in place of $|x|^{2-n}$. This derives mainly from two facts. First, the function $x \mapsto |x|^{2-n}$ is harmonic in $\mathbf{R}^n \setminus \{0\}$ and it is the fundamental solution for the Laplace equation $\Delta u = 0$. Secondly, the harmonic functions, i.e. the solutions of $\Delta u = 0$, minimize the Dirichlet integral $\int |\nabla u|^2 \, d\mathcal{L}^n$ for given boundary values. The capacity C_{n-2} is related to this variational problem since it can be shown that for compact sets $F \subset \mathbf{R}^n$,

$$C_{n-2}(F) = c(n) \inf \int |\nabla u|^2 \, d\mathcal{L}^n,$$

where the infimum is taken over all functions $u \in C_0^\infty$ with $u = 1$ on F. Many other function classes, e.g. solutions of other partial differential equations, Sobolev spaces, etc., have their natural capacities defined via a potential-theoretic approach or a minimization process. They can be used to describe removable singularities, boundary behaviour and other central properties and they are usually related to Hausdorff measures as Riesz capacities above, see e.g. Carleson [1], Evans and Gariepy [1], Hayman and Kennedy [1], Heinonen, Kilpeläinen and Martio [1], Landkof [1] and Ziemer [1].

A proof similar to that of Frostman's lemma also gives the following result on the existence of sets with finite Hausdorff measure in a set of infinite measure, see Davies [1]. For $\mathcal{F}_\sigma$ -sets this proof can be found also in Falconer [4, Theorem 5.4], for Borel (and Suslin) sets, see Federer [3, 2.10.47–48] and Rogers [1, Chapter 2.7, Theorem 57]. Below we shall give a new proof due to Howroyd which works in more general metric spaces.

8.13. Theorem. *For any Borel set $B \subset \mathbf{R}^n$,*

$$\mathcal{H}^s(B) = \sup\left\{\mathcal{H}^s(C) : C \subset B \text{ is compact with } \mathcal{H}^s(C) < \infty\right\}.$$

Theorems 8.8 and 8.13 are closely related but in some sense 8.13 seems to be stronger: using 8.13 and the density upper bound in 6.2 (1) one easily gets Theorem 8.8, but I do not know any easy way to go in the other direction.

We shall now describe the method of Howroyd [1] to prove Frostman's lemma 8.8 and Theorem 8.13 in compact metric spaces. It also works for Suslin subsets of complete separable metric spaces. One of the main tools is the weighted Hausdorff measures that we first study briefly. A more extensive and general treatment can be found in Federer [3, 2.10.24] and Kelly [1].

For the rest of this chapter X will be a compact metric space and $0 \leq s < \infty$.

Weighted Hausdorff measures

8.14. For $0 < \delta \leq \infty$ and any function $f\colon X \to [0, \infty]$ set

$$\lambda_\delta^s(f) = \inf \sum_i c_i\, d(E_i)^s$$

where the infimum is taken over all finite or countable families $\{(E_i, c_i)\}$ such that $0 < c_i < \infty$, $E_i \subset X$, $d(E_i) \leq \delta$ and

$$f \leq \sum_i c_i\, \chi_{E_i}.$$

Obviously $\lambda^s_\delta(f)$ is non-increasing in δ and we can define

$$\lambda^s(f) = \lim_{\delta\downarrow 0} \lambda^s_\delta(f).$$

For $A \subset X$ and $f = \chi_A$ we set $\lambda^s_\delta(A) = \lambda^s_\delta(\chi_A)$ and $\lambda^s(A) = \lambda^s(\chi_A)$ so that

$$\lambda^s_\delta(A) = \inf\Big\{\sum_i c_i\, d(E_i)^s : \chi_A \le \sum_i c_i\, \chi_{E_i},\, c_i > 0,\, d(E_i) \le \delta\Big\}.$$

It is easy to verify that

$$\lambda^s(f) \le \int^* f\, d\mathcal{H}^s. \tag{8.15}$$

In fact, this holds as equality, see Federer [3, 2.10.24]. We shall content ourselves with proving the following simpler special result.

8.16. Lemma. $\mathcal{H}^s(X) \le 30^s \lambda^s(X)$.

Proof. Let $\delta > 0$ and $0 < t < 1$. If c_i and E_i are such that $d(E_i) \le \delta$ and $\sum_i c_i\, \chi_{E_i} \ge 1$, we can find open balls U_i with $E_i \subset U_i$ and $d(U_i) \le 3d(E_i) \le 3\delta$. Then $\sum_i c_i\, \chi_{U_i} > t$. For $k = 1, 2, \dots,$ the sets $\{x : \sum_{i=1}^k c_i\, \chi_{U_i}(x) > t\}$ are open and their union is X. Since X is compact we can find k such that $\sum_{i=1}^k c_i\, \chi_{U_i} > t$ on X. Let B_i be a closed ball containing U_i and with $d(B_i) \le 2d(U_i)$. We have now

$$X = \Big\{x : \sum_{i=1}^k c_i\, \chi_{B_i}(x) > t\Big\}$$

and

$$\sum_{i=1}^k c_i\, d(B_i)^s \le 6^s \sum_i c_i\, d(E_i)^s.$$

Hence the lemma will follow once we have verified the following general statement.

If $0 < t < \infty$, $0 < \varepsilon \le \infty$, $c_1, \dots, c_k$ are positive numbers, and $B_1, \dots, B_k$ are closed balls with $d(B_i) \le \varepsilon$, then

$$\mathcal{H}^s_{5\varepsilon}\Big(\Big\{x : \sum_{i=1}^k c_i\, \chi_{B_i}(x) > t\Big\}\Big) \le t^{-1} 5^s \sum_{i=1}^k c_i\, d(B_i)^s. \tag{1}$$

By approximating the c_i's we may assume that each c_i is a positive rational, and then by multiplying with a common denominator we may assume that each c_i is a positive integer. Let m be the least integer with $m \geq t$. Set

$$A = \Big\{x : \sum_{i=1}^{k} c_i \, \chi_{B_i}(x) > t\Big\}.$$

Denote $\mathcal{B} = \{B_1, \ldots, B_k\}$ and define $u \colon \mathcal{B} \to \mathbf{Z}$ by $u(B_i) = c_i$; obviously we may assume $B_i \neq B_j$ for $i \neq j$ so that u is well-defined. We define by induction integer-valued functions $v_0, \ldots, v_m$ on $\mathcal{B}$ and subfamilies $\mathcal{B}_1, \ldots, \mathcal{B}_m$ of $\mathcal{B}$ starting with $v_0 = u$. Using Theorem 2.1 we find a disjoint subfamily $\mathcal{B}_1$ of $\mathcal{B}$ such that

$$A \subset \bigcup_{B \in \mathcal{B}_1} 5B.$$

Then we define inductively for $j = 1, \ldots, m$, referring again to 2.1, disjoint subfamilies $\mathcal{B}_j$ of $\mathcal{B}$ such that

$$\begin{aligned} &\mathcal{B}_j \subset \big\{B \in \mathcal{B} : v_{j-1}(B) \geq 1\big\}, \\ &A \subset \bigcup_{B \in \mathcal{B}_j} 5B \end{aligned}$$

and the functions v_j such that

$$\begin{aligned} v_j(B) &= v_{j-1}(B) - 1 \quad && \text{for } B \in \mathcal{B}_j, \\ v_j(B) &= v_{j-1}(B) && \text{for } B \in \mathcal{B} \setminus \mathcal{B}_j. \end{aligned}$$

This is possible since for $j < m$,

$$A \subset \Big\{x : \sum_{x \in B \in \mathcal{B}} v_j(B) \geq m - j\Big\},$$

whence every $x \in A$ belongs to some ball $B \in \mathcal{B}$ with $v_j(B) \geq 1$. Thus

$$\begin{aligned} m\mathcal{H}^s_{5\epsilon}(A) &\leq \sum_{j=1}^{m} \sum_{B \in \mathcal{B}_j} d(5B)^s \leq 5^s \sum_{j=1}^{m} \sum_{B \in \mathcal{B}_j} \big(v_{j-1}(B) - v_j(B)\big)\, d(B)^s \\ &\leq 5^s \sum_{B \in \mathcal{B}} \sum_{j=1}^{m} \big(v_{j-1}(B) - v_j(B)\big)\, d(B)^s \leq 5^s \sum_{B \in \mathcal{B}} u(B)\, d(B)^s. \end{aligned}$$

This gives (1) and completes the proof of the lemma. $\square$

Frostman's lemma in compact metric spaces

We now prove Frostman's lemma on X using Lemma 8.16 and the Hahn–Banach theorem.

8.17. Theorem. *Let $0 < \delta \le \infty$. There is a Radon measure μ on X such that $\mu(X) = \lambda_\delta^s(X)$ and*

(1) $$\mu(E) \le d(E)^s \quad \text{for all } E \subset X \text{ with } d(E) < \delta.$$

In particular, if $\mathcal{H}^s(X) > 0$ there exist $\delta > 0$ and μ satisfying (1) and $\mu(X) > 0$.

Proof. The last statement follows from Lemma 8.16. We define a function p on the space $C(X)$ of continuous real-valued functions on X by

$$p(f) = \inf \sum_i c_i \, d(E_i)^s$$

where the infimum is taken over all finite or countable families $\{(E_i, c_i)\}$ such that $0 < c_i < \infty$, $E_i \subset X$, $d(E_i) \le \delta$ and

$$f \le \sum_i c_i \, \chi_{E_i}.$$

For non-negative $f \in C(X)$ we have then $p(f) = \lambda_\delta^s(f)$. It is easy to verify that

$$\begin{aligned} &p(tf) = t\,p(f) && \text{for } f \in C(X) \text{ and } t \ge 0, \\ &p(f+g) \le p(f) + p(g) && \text{for } f, g \in C(X). \end{aligned}$$

By the Hahn–Banach theorem, see e.g. Rudin [2, Theorem 3.2], we can extend the linear functional $c \mapsto c\,p(1)$, $c \in \mathbf{R}$, from the subspace of constant functions to a linear functional $L \colon C(X) \to \mathbf{R}$ satisfying

$$L(1) = p(1) = \lambda_\delta^s(X)$$

and

$$-p(-f) \le L(f) \le p(f) \quad \text{for } f \in C(X).$$

If $f \ge 0$, $p(-f) = 0$ and so $L(f) \ge 0$. Hence we can use the Riesz representation theorem 1.16 to find a Radon measure μ on X such that $L(f) = \int f \, d\mu$ for $f \in C(X)$. Then also $\mu(X) = \lambda_\delta^s(X)$. If $E \subset X$

with $d(E) < \delta$ one can easily construct a non-increasing sequence of continuous functions f_i such that $0 \le f_i \le 1$, $f_i = 1$ on E and $\operatorname{spt} f_i \subset E(1/i)$. Then

$$\mu(E) \le \lim_{i\to\infty} \int f_i \, d\mu \le \lim_{i\to\infty} \lambda_\delta^s\big(E(1/i)\big) \\ \le \lim_{i\to\infty} d\big(E(1/i)\big)^s = d(E)^s.$$

This completes the proof. □

Existence of subsets with finite Hausdorff measure

8.18. "Dyadic balls". For the proof of the next theorem we introduce a very rough analogue of the family of dyadic cubes in $\mathbf{R}^n$. For each positive integer k select, using the compactness of X, a finite sequence $B_{k,1}, \dots, B_{k,m_k}$ from the family $\{U(x, 2^{-k}) : x \in X\}$ with centres $x_{k,1}, \dots, x_{k,m_k}$ such that

$$X = \bigcup_{j=1}^{m_k} U(x_{k,j}, 2^{-k-1}).$$

Let $\mathcal{B} = \{B_{k,j} : j = 1, \dots, m_k,\ k = 1, 2, \dots\}$. Obviously for each $\varepsilon > 0$ the subfamily $\{B \in \mathcal{B} : d(B) \ge \varepsilon\}$ is finite.

8.19. Theorem. *For a compact metric space X,*

$$\mathcal{H}^s(X) = \sup\big\{\mathcal{H}^s(C) : C \subset X \text{ is compact with } \mathcal{H}^s(C) < \infty\big\}.$$

Proof. We may assume $\mathcal{H}^s(X) = \infty$. Let $M < \infty$ and use Lemma 8.16 to find δ, $0 < \delta < 1$, such that $\lambda_\delta^s(X) > M$. Let $\mathcal{B}$ be the family of balls of 8.18 and let $\mathcal{M}_\delta^s$ be the set of all Radon measures μ on X for which $\mu(B) \le d(B)^s$ for all $B \in \mathcal{B}$ with $d(B) < \delta$. Put

$$h = \sup\{\mu(X) : \mu \in \mathcal{M}_\delta^s\}$$

and

$$H = \{\mu \in \mathcal{M}_\delta^s : \mu(X) = h\}.$$

By Theorem 8.17

$$M < \lambda_\delta^s(X) \le h.$$

If $2^{-k} < \delta/2$, $\mu(X) \le m_k \delta^s$ for all $\mu \in \mathcal{M}_\delta^s$. Pick $\mu_i \in \mathcal{M}_\delta^s$ with $\mu_i(X) \to h$. By Theorem 1.23 (the same proof works for compact metric spaces instead of $\mathbf{R}^n$) (μ_i) has a sub-sequence converging weakly to some Radon measure μ. Using Theorem 1.24 we find that $\mu \in \mathcal{M}_\delta^s$. Since also $\mu(X) = h$, $\mu \in H$, whence $H \neq \emptyset$. Moreover one easily verifies that H is convex and compact with respect to the weak topology. Consequently by the Krein–Milman theorem, see e.g. Rudin [2, Theorem 3.21], there is an extreme point $\mu \in H$. This implies that if $\mu = t\mu_1 + (1-t)\mu_2$, $0 < t < 1$, and $\mu_1, \mu_2 \in H$, then $\mu = \mu_1 = \mu_2$.

Let $0 < \varepsilon < \delta$ and $1 < \tau < 2$. Define

$$D_{\tau,\varepsilon} = \bigcup \{B \in \mathcal{B} : d(B) \leq \varepsilon \text{ and } \tau\mu(B) > d(B)^s\}.$$

We claim that $\mu(X \setminus D_{\tau,\varepsilon}) = 0$.

To prove this suppose $\mu(X \setminus D_{\tau,\varepsilon}) > 0$ and let $B_1, \dots, B_m$ be all the balls of $\mathcal{B}$ whose diameter is greater than ε ordered so that

$$d(B_1) \geq \dots \geq d(B_m).$$

Define inductively for $i = 1, \dots, m$,

$$\begin{aligned} &A_1 = X \setminus D_{\tau,\varepsilon}, \\ &A_{i+1} = A_i \setminus B_i \quad \text{if } \mu(A_i \setminus B_i) \geq \mu(A_i \cap B_i), \\ &A_{i+1} = A_i \cap B_i \quad \text{if } \mu(A_i \setminus B_i) < \mu(A_i \cap B_i). \end{aligned}$$

Let $A = A_{m+1}$. Then A is a Borel set, $\mu(A) > 0$, since $\mu(A_{i+1}) \geq \frac{1}{2}\mu(A_i)$ for $i = 1, \dots, m$ and $\mu(A_1) > 0$, and either $A \subset B$ or $A \cap B = \emptyset$ for every $B \in \mathcal{B}$ with $d(B) > \varepsilon$. Using the fact that $\mu(\{x\}) = 0$ for $x \in X$, as $\mu \in \mathcal{M}_\delta^s$, we find Borel sets C_1 and C_2 such that

$$A = C_1 \cup C_2, \quad C_1 \cap C_2 = \emptyset \quad \text{and} \quad \mu(C_1) = \mu(C_2) = \tfrac{1}{2}\mu(A).$$

We verify this as a lemma after the rest of the proof.

We define now Radon measures μ_1 and μ_2 by

$$\begin{aligned} \mu_1(E) &= \tau\mu(C_1 \cap E) + (2-\tau)\,\mu(C_2 \cap E) + \mu(E \setminus A), \\ \mu_2(E) &= (2-\tau)\,\mu(C_1 \cap E) + \tau\mu(C_2 \cap E) + \mu(E \setminus A) \end{aligned}$$

for $E \subset X$. Then $\mu = (\mu_1 + \mu_2)/2$, $\mu_1(X) = \mu_2(X) = \mu(X) = h$, $\mu_1(C_1) = \tau\mu(C_1) > \mu(C_1)$ and $\mu_2(C_2) = \tau\mu(C_2) > \mu(C_2)$. Thus by the extremality of μ μ_1 and μ_2 cannot both belong to $\mathcal{M}_\delta^s$. Suppose for example $\mu_1 \notin \mathcal{M}_\delta^s$ which means that there exists $B \in \mathcal{B}$ with $d(B) < \delta$ for which $\mu_1(B) > d(B)^s$. Then $d(B) \leq \varepsilon$ since otherwise either $A \subset B$

or $A \cap B = \emptyset$ and in both cases $\mu_1(B) = \mu(B) \leq d(B)^s$. Since $\tau > 1 > 2 - \tau > 0$, we have

$$\begin{aligned} \tau\mu(B) &\geq \tau\mu(C_1 \cap B) + (2-\tau)\,\mu(C_2 \cap B) + \mu(B \setminus A) \\ &= \mu_1(B) > d(B)^s, \end{aligned}$$

whence $B \subset D_{\tau,\varepsilon} \subset X \setminus A$. This gives a contradiction:

$$d(B)^s < \mu_1(B) = \mu(B \setminus A) \leq \mu(B) \leq d(B)^s.$$

Hence we have shown that $\mu(X \setminus D_{\tau,\varepsilon}) = 0$ for all $0 < \varepsilon < \delta$ and $1 < \tau < 2$.

Let p be a positive integer with $1/p < \delta$ and define

$$D = \bigcap_{j=2}^{\infty} \bigcap_{i=p}^{\infty} D_{1+1/j,1/i}.$$

Then $\mu(X \setminus D) = 0$ and for all $x \in D$

$$\lim_{\varepsilon \downarrow 0} \sup \big\{ d(B)^{-s}\mu(B) : x \in B \in \mathcal{B},\, d(B) \leq \varepsilon \big\} = 1.$$

For each $B \in \mathcal{B}$ choose a closed ball B^* with $B \subset B^*$ and $d(B^*) \leq 2d(B)$. Using Theorem 2.1 we can find for any $\varepsilon > 0$ disjoint balls $B_1, B_2, \dots \in \mathcal{B}$ such that $\mu(B_i) \geq d(B_i)^s/2$, $d(B_i) \leq \varepsilon/10$, and $D \subset \bigcup_i 5B_i^*$. Then

$$\begin{aligned} \mathcal{H}^s_\varepsilon(D) &\leq \sum_i d(5B_i^*)^s \leq 10^s \sum_i d(B_i)^s \\ &\leq 2 \cdot 10^s \sum_i \mu(B_i) \leq 2 \cdot 10^s \mu(X), \end{aligned}$$

whence $\mathcal{H}^s(D) \leq 2 \cdot 10^s \mu(X) < \infty$. On the other hand, for any sets $E_1, E_2, \dots$ covering D with $d(E_i) < \delta/8$ we can find, by the definition of $\mathcal{B}$, balls $B_1, B_2, \dots \in \mathcal{B}$ for which $E_i \subset B_i$ and $d(B_i) \leq 8d(E_i)$ which gives

$$\begin{aligned} \sum_i d(E_i)^s &\geq 8^{-s} \sum_i d(B_i)^s \geq 8^{-s} \sum_i \mu(B_i) \geq 8^{-s}\mu(X) \\ &= 8^{-s}h > 8^{-s}M. \end{aligned}$$

Thus $\mathcal{H}^s(D) \geq 8^{-s}M$. Since $M < \infty$ was arbitrary, Theorem 1.10 (1) yields

$$\sup \big\{ \mathcal{H}^s(C) : C \subset X \text{ is compact with } \mathcal{H}^s(C) < \infty \big\} = \infty,$$

as required. □

What is left is to prove the following.

8.20. Lemma. *Let μ be a Radon measure on X such that $\mu(\{x\}) = 0$ for $x \in X$. If A is a Borel subset of X and $0 < t < \mu(A)$, there is a Borel set $B \subset A$ with $\mu(B) = t$.*

Proof. Let $\mathcal{B}$ be the family of all Borel subsets B of A such that $\mu(B) \leq t$. We introduce a partial order $\subset_\mu$ on $\mathcal{B}$ (or more precisely on the set of equivalence classes where B and C are equivalent if $\mu((B\setminus C)\cup(C\setminus B)) = 0$, but we ignore this) by setting $B \subset_\mu C$ if $\mu(B \setminus C) = 0$. We want to apply Zorn's lemma to find a maximal element in $\mathcal{B}$ with respect to this order. To do this we first have to verify that if $\mathcal{C}$ is a totally ordered subfamily of $\mathcal{B}$ then $\mathcal{C}$ has an upper bound in $\mathcal{B}$. Let

$$u = \sup\{\mu(C) : C \in \mathcal{C}\}.$$

Since $\mathcal{C}$ is totally ordered we can find a sequence $C_i \in \mathcal{C}$ such that $C_1 \subset C_2 \subset \dots$ and $\mu(C_i) \to u$. Let $C_0 = \bigcup_{i=1}^\infty C_i$. Then $\mu(C_0) = u$, $C_0 \in \mathcal{B}$ and $\mu(C) \leq \mu(C_0)$ for $C \in \mathcal{C}$. If $C \in \mathcal{C}$ we have either $C \subset C_i$ for some i or $C_i \subset C$ for all i. In the first case $C \subset C_0$ and in the second $C_0 \subset C$. In both cases $\mu(C \setminus C_0) = 0$. It follows that C_0 is an upper bound for $\mathcal{C}$ with respect to $\subset_\mu$. Thus by Zorn's lemma there is a maximal element $B_0 \in \mathcal{B}$. Then $\mu(B) \leq \mu(B_0)$ for $B \in \mathcal{B}$. We have for μ almost all $x \in A \setminus B_0$ that $\mu\big((A \setminus B_0) \cap B(x,r)\big)$ is positive for all $r > 0$ and tends to zero as $r \downarrow 0$. From this we see immediately that B_0 cannot be maximal unless $\mu(B_0) = t$. This proves the lemma. □

8.21. Remark. Joyce and Preiss [1] have recently proved that Theorem 8.13 holds for packing measures in place of Hausdorff measures. Their method also works in general metric spaces.

Exercises.

1. Show that for $A \subset \mathbf{R}^n$, $C_s(A) > 0$ if and only if there is $\mu \in \mathcal{M}(A)$ such that the potential $x \mapsto \int |x-y|^{-s}\, d\mu y$ is bounded in $\mathbf{R}^n$.
2. Show that for $A \subset \mathbf{R}^n$,
$$C_s(A) = \sup\{C_s(K) : K \subset A \text{ is compact}\}.$$
3. Show that if $A_1 \subset A_2 \subset \dots \subset \mathbf{R}^n$, then
$$C_s\Big(\bigcup_{i=1}^\infty A_i\Big) = \lim_{i\to\infty} C_s(A_i).$$
4. Give a condition for a non-decreasing function $h\colon [0,\infty) \to [0,\infty)$ which guarantees that $C_s(A) = 0$ implies $\Lambda_h(A) = 0$ for Borel sets $A \subset \mathbf{R}^n$ generalizing Theorem 8.9 (2).

5. Prove the inequality (3) in Theorem 8.10. Hint: You may use Exercises 5.5–6.
6. Use Frostman's lemma to prove that $\mathcal{H}^s(C(\lambda)) > 0$ where $C(\lambda)$ is as in 4.10 and $s = \log 2/\log(1/\lambda)$.
7. Prove inequality (8.15).